Curriculum vitæ

Personal informations

Schmitt Sylvain

4th of August 1993, Rennes, France

UPR Forêts et Sociétés, CIRAD Baillarguet

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Website

Google scholar profile

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Research activities

Overview. The primary objective of my research is to understand and model the fate of tropical forests in the face of global change and increasing anthropic disturbances, from the individual tree to the forest basin, in order to improve our ability to manage these ecosystems and predict their responses. I use several complementary and mutually informative approaches - functional & community ecology, remote sensing & climate projections - based on advanced modeling and simulations of tropical forests at multiple spatial and temporal scales.

Historical and future climates. My research activities start with climate data on multiple temporal and spatial scales and from multiple sources, from local weather stations and eddy flux towers to global remote sensing products via regional and global reanalyses, which I combine with multiple dynamic projections from global and regional climate models developed by the climate model intercomparison projects of the IPCC. I am currently working on the development of an automated tool for the statistical downscaling of dynamic regional climate projections on a national and territorial scale, based on multiple climate references, in order to better support applied sciences such as ecology and forestry, which need projections on such a fine scale.

Local forest dynamics. I am assimilating repeated field surveys, eddy flux tower measurements, local drone surveys and remotely sensed forest measurements to calibrate and evaluate the *TROLL* individual-based and spatially explicit forest simulator for jointly simulating biodiversity, carbon and water dynamics in forest ecosystems. I would therefore like to take advantage of explicit species modeling that takes into account the immense biodiversity of tropical forests and individual modeling to project the joint effects of climate change and forest management practices on the future trajectories of the different types of tropical forests and the ecosystem services they provide.

Forests in the landscapes. My research activities also include the link between landscape trajectories and regional climate in order to support projects using territories as levers for sustainable development in Amazonia. In particular, I am trying to establish a link between changes in forest cover and structure, *i.e.* degradation, deforestation and regeneration, and variations in climate, as well as the effects of climate variations on forest functioning. I study climate trends as well as extreme events and changes in fire risk in order to produce indices and projections.

The Amazon basin. Beyond understanding the local dynamics of forests and their dynamics within the landscape, I would like to study the dynamics of tropical forests in the Amazon basin as a whole in relation to the bi-stability of ecosystems, in particular between tropical moist forests and savannas, in order to better characterize potential tipping points in Amazonia, focusing on the little-studied role of biodiversity in stability and on a better spatio-temporal characterisation of possible forest trajectories.

Experience

2023 - present	**Researcher: Tropical forest Ecology and Climate Change** Research Unit Forest & Societies (CIRAD, France). *Ecology and Climate Change*
2023 *6 months*	**Postdoctoral researcher: Tropical forest modeling with joint effect of climate change and anthropic disturbances** Laboratory Botany and Modeling of Plant and Vegetation Architecture (AMAP, France). Supervisor: Dr. Isabelle Maréchaux (INRAE) *Individual-based modeling of tropical forest dynamics (TROLL)*
2021-2023 *26 months*	**Postdoctoral researcher: Mutation in the tropical tree canopy** Laboratory Ecology of Guiana Forests (ECOFOG, French Guiana) Supervisor: Drs. Niklas Tysklind & Myriam Heuertz (INRAE) *Genomic of somatic mutation detection in tropical tree crowns*
2017-2020 *39 months*	**PhD thesis: Ecological genomics of niche exploitation and individual performance in tropical forest trees** Laboratories Biodiversity Genes and Communities (BIOGECO, France) and Ecology of Guiana Forests (ECOFOG, French Guiana) Supervisor: Drs. Myriam Heuertz (INRAE) & Bruno Hérault (CIRAD) *Quantitative and ecological genomics, Functional and community ecology, Forest demography and dynamics.*
2017 *6 months*	**MSc research project: Biodiversity and resilience of tropical forest ecosystem after disturbance** Laboratory Ecology of Guiana Forests (ECOFOG, French Guiana) Supervisor: Drs. Bruno Hérault (CIRAD) & Stéphane Traissac (APT) *Individual-based modeling of tropical forest dynamics (TROLL)*
2016 *6 months*	**Research internship: Wood and leaf strategy in a low elevation rainforest of Western Ghats (India)** French Institute of Pondicherry (FIP, India) Supervisor: Drs. François Munoz (U. Grenoble) & Maxime Réjou-Méchain (IRD) *Field sampling in remote tropical forests, Measurement of functional traits, Statistics, Community and functional ecology.*
2015-2016 *6 months*	**Research internship: Species distribution modeling (SSDM)** New Caledonian Agronomic Institute (IAC, New Caledonia) Supervisor: Drs. Phillipe Birnbaum (CIRAD) & Robin Pouteau (IRD) *R package development & species distribution modeling*
2015 *3 months*	**Research internship in soil ecology** Helmholtz Zentrum München (HZM, Germany). Supervisor: Drs. René Kerner & Karin Pritsch (HZM).

Education

2017-2020 **PhD in Ecology and Evolution**, University of Bordeaux
Thesis written in English.
Dissertation committee: Pr. Xavier Vekemans, Dr. Tamara Münkemüller, Dr. Olivier Hardy, Dr. Céline Teplitsky, Dr. Caroline Scotti-Saintagne, Dr. Marta Benito-Garzón.

2015-2016 **M. Sc. in Ecology, Biodiversity and Evolution**, University of Montpellier
Grade rank: 1st out of 10.

2013-2016 **Engineer degree in natural land management**, AgroParisTech Nancy
One of France's leading schools in environmental sciences and engineering, to which entrance is based on a national competitive exam. An engineer degree is equivalent to an MSc.

2011-2013 **Undergraduate studies, majors in Biology, Chemistry, Earth Sciences and Mathematics**, Lycée François René de Chateaubriand
2-year intensive undergraduate study (Classes préparatoires), in preparation of the highly competitive entry exams to France's leading schools of science and engineering (Grandes Écoles) equivalent to years 1 and 2 of a BSc.

Publications

number of citations according to Google Scholar

16. R Gargiulo, V Decroocq, SC González-Martínez, I Paz-Vinas, JM Aury, I Lesur Kupin, C Plomion, **S Schmitt**, I Scotti, M Heuertz. Estimation of contemporary effective population size in plant populations: limitations of genomic datasets. *Evolutionary Applications*. Citation: 2. https://doi.org/10.1111/eva.13691

15. **S Schmitt**, P Heuret, V Troispoux, M Beraud, J Cazal, E Chancerel, C Cravero, E Guichoux, O Lepais, J Loureiro, W Marande, O Martin, G Vincent, J Chave, C Plomion, T Leroy, M Heuertz, N Tysklind. Low-frequency somatic mutations are heritable in plants. *PNAS.* Citation: 5. https://doi.org/10.1073/pnas.2313312121

14. V Badouard, **S Schmitt**, G Salzet, T Gaquiere, O Brunaux, C Bedeau, M Rojat, G Derroire. LoggingLab: an R package for simulating reduced impact logging in tropical forests using forest inventory data. *Ecological Modelling* https://doi.org/10.1016/j.ecolmodel.2023.110539.

13. **S Schmitt**, B Hérault, G Derroire. High intraspecific growth variability despite strong evolutionary heritage in a neotropical forest. *Ecology Letters* https://doi.org/10.1111/ele.14318.

12. **S Schmitt**, G Salzet, I Maréchaux, F Jörg Fischer, J Chave. rcontroll: an R interface for the individual-based forest dynamics simulator TROLL. *Methods in Ecology and Evolution* https://doi.org/10.1111/2041-210X.14215.

11. **S Schmitt**, M Boisseaux. Robust assessment reveals high intraspecific variability in water and carbon related leaf traits in ten neotropical tree species. *Annals of Botany* https://doi.org/10.1093/aob/mcad042. Citations: 2.

10. **S Schmitt**, T Leroy, M Heuertz, N Tysklind. Somatic mutation detection in plants: a critical evaluation through simulations and reanalyses. *Peer Community Journal* https://doi.org/10.1101/2021.10.11.462798. Citations: 6.

9. **S Schmitt**, S Trueba, S Coste, É Ducouret, N Tysklind, M Heuertz, D Bonal, B Burban, B Hérault, G Derroire (2022). Seasonal variation of leaf thickness: An overlooked component of functional trait variability. *Plant Biology* 24:458-463. https://doi.org/10.1111/plb.13395. Citations: 6.

8. **S Schmitt**, N Tysklind, M Heuertz, B Hérault (2022). Selection in space and time: Individual tree growth is adapted to tropical forest gap dynamics. From the Cover, *Molecular Ecology*, [in press]. https://doi.org/10.1111/mec.16392. Citations: 6.

7. **S Schmitt**, N Tysklind, B Hérault, M Heuertz (2021). Topography drives microgeographic adaptations of closely related species in two tropical tree species complexes. *Molecular Ecology* 30:5080-5093. https://doi.org/10.1111/mec.16116. Citations: 15.

6. **S Schmitt**, V Raevel, M Réjou‐Méchain, N Ayyappan, N Balachandran, N Barathan, G Rajashekar, F Munoz (2021). Canopy and understorey tree guilds respond differently to the environment in an Indian rain forest. *Journal of Vegetation Science* 32:e13075. https://doi.org/10.1111/jvs.13075. Citations: 2.

5. **S Schmitt**, N Tysklind, G Derroire, M Heuertz, B Hérault (2021). Topography shapes the local coexistence of tree species within species complexes of Neotropical forests. *Oecologia* 196:389-398. https://doi.org/10.1007/s00442-021-04939-2. Citations: 13.

4. **S Schmitt**, B Hérault, É Ducouret, A Baranger, N Tysklind, M Heuertz, É Marcon, SO Cazal, G Derroire (2020). Topography consistently drives intra‐and inter‐specific leaf trait variation within tree species complexes in a Neotropical forest. *Oikos* 196:1521-1530. https://doi.org/10.1111/oik.07488. Citations: 35.

3. **S Schmitt**, I Maréchaux, J Chave, F Jörg Fischer, C Piponiot, S Traissac, B Hérault (2020). Functional diversity improves tropical forest resilience: Insights from a long‐term virtual experiment. *Journal of Ecology* 108:831-843. https://doi.org/10.1111/1365-2745.13320. Citations: 53.

2. J Kattge, G Bönisch, …, **S Schmitt**, *et al.* (2020). TRY plant trait database–enhanced coverage and open access. *Global Change Biology* 26:119-188. https://doi.org/10.1111/gcb.14904. Citations: 1171.

1. **S Schmitt**, R Pouteau, D Justeau, F De Boissieu, P Birnbaum (2017). SSDM: An R package to predict distribution of species richness and composition based on stacked species distribution models. *Methods in Ecology and Evolution* 8:1795-1803. https://doi.org/10.1111/2041-210X.12841. Citations: 152.

Grants

2020	**Outgoing mobility grant.** 6-months mobility to Ivory Coast funded by the Laboratory of Excellence on Continental To Coastal Ecosystems: evolution, adaptability and governance (COTE).
2019	**International mobility support.** To participate in a conference and field work in Ecuador funded by the Univ. Bordeaux "Sciences and environments" doctoral school.
2019	**Congress Grants for Doctoral Students.** To participate in the 56th Annual meeting of the Association for Tropical Biology and Conservation funded by the Department of Environmental Sciences of the University of Bordeaux.
2017-2020	**PhD grand (three-year salary).** Funded by the French ministry of higher education and research.

Major communications *including only international conferences*

3. **S Schmitt**, R Pouteau, D Justeau, F De Boissieu, P Birnbaum (2017). SSDM: An A package to predict distribution of species richness and composition based on stacked species distribution models. *Poster at the international biogeography society 2019 Humboldt Meeting*.

2. **S Schmitt**, B Hérault, É Ducouret, A Baranger, N Tysklind, M Heuertz, É Marcon, SO Cazal, G Derroire (2019). Ontogeny and abiotic environment drive intraspecific trait variation in Neotropical tree species. *Talk at the Annual meeting of the Association for Tropical Biology and Conservation*.

1. **S Schmitt**, N Tysklind, B Hérault, M Heuertz (2018). Ecological genomics of niche exploitation and individual performance in tropical forest trees. *Poster at the European Conference of Tropical Ecology*.

Teaching, student supervision, and scientific reviewing

Student supervision:

2024. A Bourgois (6 months, co-supervisor). Simulation of regeneration in tropical forests. *Msc thesis at the University of Montpellier*.

2022. B Lagrange, F Kratz (4 months). Simulation of logging damage in TROLL. *AgroParisTech MSc project*.

2022. S Cassin, S Hoppe (4 months). Regeneration of seedlings in Paracou. *AgroParisTech MSc project*.

2022. M Lebrun, G Pouthé, Y Saliou, L Valade (2 weeks). Forest regeneration in a tropical environment. *AgroParisTech MSc project*.

2021. L Alvergnat, M Bentkowski, P Desvéronnières, O Girard Reydet (2 weeks). Functional traits of tropical trees : Inter- and intra-specific variability at the Paracou site. *AgroParisTech MSc project*.

2019. A Baranger (3 months). Modeling intraspecific variability among two species complexes from French Guiana. *AgroParisTech voluntary internship*.

2019. A Adam, A Benfredj Zaleski, N Faucherre, D Zipper (2 weeks). Variability of functional and morphological traits within the *Parvifolia* species complex. *AgroParisTech MSc project*.

2018. N Page (6 months, co-supervisor). Simulation of harvested forests with the TROLL model. *Msc thesis at the University of the French West Indies*.

2018. É Ducouret (6 months). Modeling intraspecific variability among two species complexes from French Guiana. *Msc thesis at the University of the French West Indies*.

Teaching and workshops:

2022. rcontroll: an R interface for the individual-based forest simulator TROLL (2h). *Peer workshop*.

2022. Reproducible analyses with Snakemake & Singularity (4h). *Peer workshop*.

2021-2022. Modeling diversity (4x1h). *MSc tropical rainforest module*.

2020. Reproducible analyses with Rmarkdown and Bookdown (2x1h). *Peer workshop.*

2019-2020. 3 peer workshops on Bayesian modeling (3x1h): *Animal model*, *Dirichlet Multinomial*, & *Neighbourhood models and Segmentation*

2018. Introduction to Bayesian modeling with stan (4h). *Peer workshop*.

2018-2019. Introduction to Bayesian modeling with winbugs (2x6h). *MSc Tropical Forest Ecology*.

Scientific reviewing:

(2) Biological Conservation, (2) Molecular Ecology, (2) Tree Physiology, (1) Ecological Informatics, (1) Functional Ecology. View my Publons profile.

Technical skills

Languages	French (mother tongue), English (fluent; TOEIC: 960/990), German (basic)
Field	Tree climbing, Tree sampling from the ground (Big Shot), Functional trait measurements, Permanent sample plots, Organising field missions in remote stations
Laboratory	DNA extraction, Genomic library preparation, Gene capture experiment
Statistics	Bayesian hierarchical modeling, Machine learning, Advanced statistical analyses
Informatic	Programming (R, C++, Bash, Python, Perl), R package development, Bioinformatics pipeline development (Snakemake, Singularity), High performance computing, Versioning (Git, GitHub), Reproducibility (R Markdown, Quarto, renv, Docker)